**Training Output:**

```
-----------------------------------------
| time/                |            |
|    fps               | 412        |
|    iterations        | 304        |
|    time_elapsed      | 6031       |
|    total_timesteps   | 2490368    |
| train/               |            |
|    approx_kl         | 0.0057996577 |
|    clip_fraction     | 0.0967     |
|    clip_range        | 0.15       |
|    entropy_loss      | -0.941     |
|    explained_variance | 0.81      |
|    learning_rate     | 1.06e-05   |
|    loss              | -0.0476    |
|    n_updates         | 3030       |
|    policy_gradient_loss | -0.0063  |
|    value_loss        | 0.129      |
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| time/                |            |
|    fps               | 413        |
|    iterations        | 305        |
|    time_elapsed      | 6045       |
|    total_timesteps   | 2498560    |
| train/               |            |
|    approx_kl         | 0.005631931 |
|    clip_fraction     | 0.0979     |
|    clip_range        | 0.15       |
|    entropy_loss      | -0.94      |
|    explained_variance | 0.824     |
|    learning_rate     | 1.03e-05   |
|    loss              | -0.054     |
|    n_updates         | 3040       |
|    policy_gradient_loss | -0.00711 |
|    value_loss        | 0.107      |
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| time/                |            |
|    fps               | 413        |
|    iterations        | 306        |
|    time_elapsed      | 6058       |
|    total_timesteps   | 2506752    |
| train/               |            |
|    approx_kl         | 0.0059254654 |
|    clip_fraction     | 0.101      |
|    clip_range        | 0.15       |
|    entropy_loss      | -0.934     |
```

| explained_variance | 0.864 |
| learning_rate | 1.01e-05 |
| loss | -0.032 |
| n_updates | 3050 |
| policy_gradient_loss | -0.00553 |
| value_loss | 0.131 |
---
100% ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2,506,752/2,500,000 [ 1:40:49 < 0:00:00 , 603 it/s ]
100% ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2,506,752/2,500,000 [ 1:40:49 < 0:00:00 , 693 it/s ]
[INFO ] 19:47:13 __main__ Training complete.

**Backtest Result:**

[INFO ] 19:47:15 engine.backtester OOS Test: Simulating 6,792 bars on EURUSD...
[INFO ] 19:47:31 engine.backtester OOS Test complete. Final balance: $7,351.04 | Trades executed: 408

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QUANT PREDATOR v2 — OUT-OF-SAMPLE
═══════════════════════════════════════════════════════════
Starting Balance : $ 10,000.00
Ending Balance : $ 7,351.04
Total Return : -26.49%
Annual Return : -25.02%
─────────────────────────────────────────────────────────
Max Drawdown : -33.15%
Sharpe Ratio : -0.682 (H1-annualised)
Sortino Ratio : -0.844
Calmar Ratio : -0.755
─────────────────────────────────────────────────────────
Total Trades : 307
Win Rate : 25.7%
Profit Factor : 0.860
Avg Win : $ +205.50
Avg Loss : $ -82.82
Expectancy : $ -8.63
─────────────────────────────────────────────────────────
Verdict: FAILING — Not suitable for deployment.
═══════════════════════════════════════════════════════════

[INFO ] 19:47:31 __main__ TRADE FREQUENCY: 1.1 trades/day (22 bars/trade) | Target: 2-3/day (8-12 bars/trade)
[INFO ] 19:47:31 __main__ Frequency verdict: ⚠️ Too infrequent
[WARNING ] 19:47:31 __main__ VALIDATION FAILED ❌ Sharpe: -0.68 | MaxDD: 33.2%. Try best checkpoint at D:\Softwares\GitHub\quant-predator-v2\models\eurusd\checkpoints/best_model.zip, or

run engine/optuna_search.py for hyperparameter tuning.

**Wandb Run Summary:**

Run summary:
wandb: oos/bars_per_trade 22.12378
wandb: oos/expectancy_usd -8.62844
wandb: oos/max_drawdown -33.15026
wandb: oos/n_trades 307
wandb: oos/profit_factor 0.85972
wandb: oos/sharpe_ratio -0.68214
wandb: oos/total_return -26.48958
wandb: oos/trades_per_day 1.08481
wandb: oos/validation_passed 0
wandb: oos/win_rate 25.7329